选：DDBCB

考试科目：高等数学(上) A　　2022–2023学年秋季学期期末考试试卷

南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目：高等数学(上) A　　开课单位：数学系

考试时长：120 分钟　　命题教师：

题号	1	2	3	4	5	6	7	8
分值	20 分	20 分	8 分	10 分	7 分	10 分	16 分	9 分

本试卷共 8 道大题, 满分 100 分. (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

注意：本试卷里的中文为直译（即完全按英文字面意思直接翻译），所有数学词汇的定义请参照教材（Thomas' Calculus，13th Edition）中的定义。如果其中有些数学词汇的定义不同于中文书籍（比方说同济大学的高等数学教材）里的定义，以教材（Thomas' Calculus，13th Edition）中的定义为准。

1. (20pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

(1) If $f(x)=\frac{ax+b}{x^2-1}$ has a local extreme of 1 at $x=3$, then

(A) $a=3,\ b=-1$.　　(B) $a=4,\ b=-4$.

(C) $a=5,\ b=-7$.　　(D) $a=6,\ b=-10$.

(2) Let $f(x)=\dfrac{\tan x}{|x|\,(x-\frac{\pi}{2})^4}$. Which of the following statements must be **correct**?

(A) f is continuous at $x=0$ and f has a jump discontinuity at $x=\frac{\pi}{2}$.

(B) f has a jump discontinuity at $x=0$ and f is continuous at at $x=\frac{\pi}{2}$.

(C) f has an infinite discontinuity at $x=0$ and f has an oscillating discontinuity at $x=\frac{\pi}{2}$.

(D) f has a jump discontinuity at $x=0$ and f has an infinite discontinuity at $x=\frac{\pi}{2}$.

(3) The number of asymptotes of $y=e^{\frac{1}{x^2}}\arctan\dfrac{x^2+x+1}{(x-1)(x+2)}$ is

(A) 1.　　(B) 2.

(C) 3.　　(D) 4.

(4) Which of the following improper integrals is divergent ?

(A) $\int_{-1}^{1}\ln|x|\,dx$.　　(B) $\int_0^1\frac{e^x}{\sqrt{1-x}}\,dx$.

(C) $\int_0^{+\infty}\frac{1}{x^2}\,dx$.　　(D) $\int_e^{+\infty}\frac{1}{x\ln^2 x}\,dx$.

(5) Suppose that $a<0<b$, and $f(x)$ is continuous on (a,b). Let

$$F(x)=\begin{cases}\dfrac{\int_0^x t\,f(t)\,dt}{x}, & x\neq 0\\ 0, & x=0\end{cases}$$

Which of the following statements must be **correct**?

(A) F is differentiable on (a,b) and F' is not continuous at $x=0$.

(B) F is differentiable on (a,b) and F' is continuous at $x=0$.

(C) F is not differentiable on (a,b) and F is continuous at $x=0$.

(D) None of the above statements is correct.

2. (20 pts) Fill in the blanks.

(1) The number of the real roots for the equation $x^3-4x^2+x+1=0$ is ____3____.

(2) If $f(x)=(1+x)(1+2x)\cdots(1+10x)$, then $f'(0)=$ ____55____.

(3) Use Euler's method to find the approximation for the solution of

$$y'=1+xy, \qquad y(0)=1.$$

Take $dx=0.5$, and start at $x_0=0, y_0=1$. Then $y_2=$ _____.

(4) If $f(x)=\arctan\frac{1+x}{1-x}$, then $f'(0)=$ ____1____.

(5) If $\int_0^1 \frac{e^x}{x+1}dx=a$, then $\int_0^1 \frac{e^x}{(x+1)^2}dx=$ _____.

3. (8 pts) Solve the following first-order linear differential equation

$$xy'+2y=x^2+1, \qquad x>0.$$

4. (10 pts)

(1) Find the area of the region enclosed by the curves $y=x^2-2x$, $y=0$, $x=1$, and $x=3$.

(2) Find the volume of the solid generated by revolving the region in (1) about the y-axis.

5. (7 pts) Assume that f is differentiable at $x=1$, and $f(1)=1$, $f'(1)=2$. Find the value of $\lim_{n\to\infty}\left(f\left(1+\frac{1}{n}\right)\right)^n$.

6. (10 pts) Evaluate the following limits.

(1) $\lim_{x\to 0^+}\frac{\ln\tan 7x}{\ln\tan 2x}$.

(2) $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k}{n^2}\sin^2\left(1+\frac{k}{n}\right)$.

7. (16 pts) Evaluate the integrals.

(1) $\int x\tan^2 x\,dx$.

(2) $\int_0^{+\infty}\frac{\tan^{-1}e^x}{e^{2x}}dx$.

(3) $\int_1^{\sqrt{3}}\frac{1}{x(1+x^2)^2}dx$.

3.

$$xy' + 2y = x^2+1 \qquad (x>0)$$

$$y' + \frac{2}{x}y = x + \frac{1}{x} \qquad \left(U(x) = e^{\int \frac{2}{x}dx} = x^2\right)$$

$$x^2y' + 2xy = x^3 + x$$

$$(x^2y)' = x^3 + x$$

$$x^2y = \frac{1}{4}x^4 + \frac{1}{2}x^2 + C$$

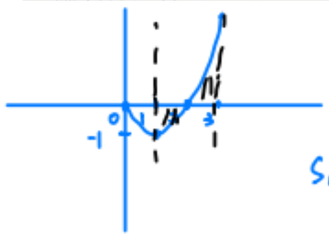
$$y = \frac{1}{4}x^2 + \frac{1}{2} + \frac{C}{x^2}$$

4. (10 pts)

(1) Find the area of the region enclosed by the curves $y = x^2 - 2x$, $y = 0$, $x = 1$, and $x = 3$.

(2) Find the volume of the solid generated by revolving the region in (1) about the y-axis.

5. (7 pts) Assume that f is differentiable at $x = 1$, and $f(1) = 1$, $f'(1) = 2$. Find the value of $\lim_{n\to\infty}\left(f\left(1+\frac{1}{n}\right)\right)^n$.



(1).

由图可知，该面积被分为3部分

$$S_1 = \left|\int_1^2 (x^2-2x)dx\right| = \left[x^3 - \frac{1}{3}x^3\right]\Big|_1^2 = 4 - \frac{8}{3} - 1 + \frac{1}{3} = 3 - \frac{7}{3} = \frac{2}{3}$$

$$S_2 = \int_2^3 (x^2-2x)dx = \left[\frac{1}{3}x^3 - x^2\right]\Big|_2^3 = 9 - 9 - \frac{8}{3} + 4 = \frac{4}{3}$$

故总面积 $S = S_1 + S_2 = 2$

(2).

$$V_1 = \left|\int_1^2 2\pi x(x^2-2x)\,dx\right| = 2\pi\int_1^2 (2x^2-x^3)dx = 2\pi\left[\frac{2}{3}x^3 - \frac{1}{4}x^4\right]\Big|_1^2 = 2\pi\left[\left(\frac{16}{3}-4\right) - \left(\frac{2}{3}-\frac{1}{4}\right)\right] = 2\pi\left[\frac{4}{3} - \frac{2}{3} + \frac{1}{4}\right] = \frac{11}{6}\pi$$

$$V_2 = \int_2^3 2\pi x(x^2-2x)\,dx = 2\pi\left[\frac{1}{4}x^4 - \frac{2}{3}x^3\right]\Big|_2^3 = 2\pi\left[\left(\frac{81}{4}-\frac{54}{3}\right) - \left(4-\frac{16}{3}\right)\right] = 2\pi\left[\frac{81}{4}\cdot\frac{16}{4} + \frac{16-54}{3}\right] = 2\pi\left[\frac{65}{4} - \frac{38}{3}\right] = 2\pi\,\frac{195-152}{12} = \frac{43}{6}\pi$$

故总体积 $V = V_1 + V_2 = 9\pi$

5. (7 pts) Assume that f is differentiable at $x = 1$, and $f(1) = 1$, $f'(1) = 2$. Find the value of $\lim_{n\to\infty}\left(f\left(1+\frac{1}{n}\right)\right)^n$.

5. 分析：$(f(1))^n = 1^\infty$ 型未定式：凑 $(1+口)^{\frac{1}{口}}$

解：$$\lim_{n\to\infty}\left(f\left(1+\frac{1}{n}\right)\right)^n = \lim_{n\to\infty}\left(1 + f\left(1+\frac{1}{n}\right) - 1\right)^{\frac{1}{f(1+\frac{1}{n})-1}\cdot\frac{f(1+\frac{1}{n})-1}{\frac{1}{n}}}$$

$$\lim_{n\to\infty}\left(1 + f\left(1+\frac{1}{n}\right) - 1\right)^{\frac{1}{f(1+\frac{1}{n})-1}} = e$$

因为 $f'(1) = 2$ 即 $\lim_{t\to 0}\frac{f(1+t)-f(1)}{t} = 2$

$$\lim_{n\to\infty}\frac{f\left(\frac{1}{n}+1\right)-f(1)}{\frac{1}{n}} \xlongequal{令 t=\frac{1}{n}} \lim_{t\to 0^+}\frac{f(1+t)-f(1)}{t} = 2$$

故由极限运算性质知：

$$\lim_{n\to\infty}\left(f\left(1+\frac{1}{n}\right)\right)^n = e^2$$

6. (10 pts) Evaluate the following limits.

(1) $\lim_{x\to 0^+}\dfrac{\ln\tan 7x}{\ln\tan 2x}$.

(2) $\lim_{n\to\infty}\sum_{k=1}^{n}\dfrac{k}{n^2}\sin^2\left(1+\dfrac{k}{n}\right)$.

6.

(1). $$\lim_{x\to 0^+}\frac{\ln(\tan 7x)}{\ln(\tan 2x)} \xlongequal{洛} \lim_{x\to 0^+}\frac{\frac{7\sec^2 7x}{\tan 7x}}{\frac{2\sec^2 2x}{\tan 2x}} = \lim_{x\to 0^+}\left(\frac{7}{2}\cdot\frac{\sec^2 7x}{\sec^2 2x}\cdot\frac{\sin 2x}{\cos 2x}\cdot\frac{\cos 7x}{\sin 7x}\right) = \lim_{x\to 0^+}\left(\frac{\cos 2x}{\cos 7x}\cdot\frac{7}{2}\cdot\frac{\sin 2x}{\sin 7x}\right)$$

$$\xlongequal{x\to 0 时, \sin x\sim x} \lim_{x\to 0^+}\left(\frac{\cos 2x}{\cos 7x}\cdot\frac{7}{2}\cdot\frac{2x}{7x}\right) = 1$$

(2). $$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k}{n^2}\sin^2\left(1+\frac{k}{n}\right) = \lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{k}{n}\sin^2\left(1+\frac{k}{n}\right) = \int_0^1 x\sin^2(1+x)\,dx$$

分析：反对幂指三，把谁放微分里，分步之后就ok了。

找 $\sin^2(1+x)$ 原函数（草稿纸上完成）$\int\sin^2(1+x)dx = \int\frac{1-\cos(2+2x)}{2}dx = \frac{1}{2}x - \frac{1}{4}\sin(2x+2)$

$$= \int_0^1 x\,d\left[\frac{1}{2}x - \frac{1}{4}\sin(2x+2)\right] = x\left[\frac{1}{2}x - \frac{1}{4}\sin(2x+2)\right]\Big|_0^1 - \int_0^1\frac{1}{2}x\,dx + \int_0^1\frac{1}{4}\sin(2x+2)dx = \frac{1}{2} - \frac{1}{4}\sin 4 - \frac{1}{4}x^2\Big|_0^1 - \frac{1}{8}\cos(2x+2)\Big|_0^1 = \frac{1}{2} - \frac{1}{4}\sin 4 - \frac{1}{4} - \frac{1}{8}\cos 4 + \frac{1}{8}\cos 2 = \frac{1}{4} - \frac{1}{4}\sin 4 - \frac{1}{8}\cos 4 + \frac{1}{8}\cos 2$$

7. (16 pts) Evaluate the integrals.

(1) $\int x\tan^2 x\,dx.$

(2) $\int_0^{+\infty} \frac{\tan^{-1} e^x}{e^{2x}}\,dx.$

(3) $\int_1^{\sqrt{3}} \frac{1}{x(1+x^2)^2}\,dx.$

(4). $\int \frac{\ln(1-x^2)}{x^2\sqrt{1-x^2}}\,dx$

7.(1). 同上题 思路

解: $\int x(\sec^2 x\cdot 1)\,dx = \int x\,d(\tan x) - \int x\,dx = x\tan x - \int \tan x\,dx - \frac{1}{2}x^2 = x\tan x - \frac{1}{2}x^2 - \ln|\sec x| + C$

(2). $\int_0^{+\infty} \frac{\arctan(e^x)}{e^{2x}}\,dx$ 该积分为无穷区间上的反常积分. 无瑕点.

$= \int_0^{+\infty} \frac{\arctan(e^x)}{e^{3x}}\,de^x \xlongequal{\text{令 } t=e^x} \int_1^{+\infty} \frac{\arctan t}{t^3}\,dt$ (反对幂三指)

$= -\frac{1}{2}\int_1^{+\infty} \arctan t\,d\left(\frac{1}{t^2}\right) = -\frac{1}{2}\frac{\arctan t}{t^2}\Big|_1^{+\infty} + \frac{1}{2}\int_1^{+\infty}\frac{1}{t^2}\cdot\frac{1}{(t^2+1)}\,dt = -\frac{1}{2}\left[\lim_{\alpha\to+\infty}\frac{\arctan\alpha}{\alpha^2} - \frac{\frac{\pi}{4}}{1}\right] + \frac{1}{2}\int_1^{+\infty}\frac{1}{t^2}\,dt - \frac{1}{2}\int_1^{+\infty}\frac{1}{t^2+1}\,dt$

$= -\frac{1}{2}\left[0-\frac{\pi}{4}\right] - \frac{1}{2}\left(-\frac{1}{t} + \arctan t\right)\Big|_1^{+\infty} = \frac{\pi}{8} - \frac{1}{2}\left(\lim_{\alpha\to+\infty}\frac{1}{\alpha} + \arctan\alpha - 1 - \frac{\pi}{4}\right) = \frac{\pi}{8} - \frac{1}{2}\left(\frac{\pi}{2} - \frac{\pi}{4} - 1\right) = \frac{1}{2}$

(3). 令 $x=\tan t$ $\quad \frac{\pi}{4}\le t\le \frac{\pi}{3}$

$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{\tan t(\sec^4 t)}\sec^2 t\,dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos^3 t}{\sin t}\,dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1-\sin^2 t}{\sin t}\,d(\sin t) \xlongequal{\text{令 } u=\sin t} \int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \frac{1}{u}\,du - \int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} u\,du = \ln\frac{\sqrt{3}}{\sqrt{2}} - \frac{1}{2}(u^2)\Big|_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}}$

$= \frac{1}{2}\ln\frac{3}{2} - \frac{1}{2}\left(\frac{3}{4}-\frac{2}{4}\right) = \frac{1}{2}\ln\frac{3}{2} - \frac{1}{8}$

(4). 思路: $[\ln(1-x^2)]' = \frac{-2x}{1-x^2}$ 是幂函数 —— 分步积分即可使难积的 $\ln(1-x^2)$ 消失变成幂函数

—— 找 $\frac{1}{x^2\sqrt{1-x^2}}$ 的原函数.



草稿纸: $\int \frac{1}{x^2\sqrt{1-x^2}}\,dx \xlongequal{x=\sin t} \int \frac{1}{\sin^2 t\cos t}\cos t\,dt = \int \csc^2 t\,dt = -\frac{\cos t}{\sin t} = -\frac{\sqrt{1-x^2}}{x}$

解: $\int \frac{\ln(1-x^2)}{x^2\sqrt{1-x^2}}\,dx = -\int \ln(1-x^2)\,d\left(\frac{\sqrt{1-x^2}}{x}\right) = -\ln(1-x^2)\cdot\frac{\sqrt{1-x^2}}{x} + \int \frac{\sqrt{1-x^2}}{x}\cdot\frac{-2x}{1-x^2}\,dx$

$= -\frac{\sqrt{1-x^2}}{x}\cdot\ln(1-x^2) - 2\int \frac{1}{\sqrt{1-x^2}}\,dx = -\frac{\sqrt{1-x^2}}{x}\ln(1-x^2) - 2\arcsin x + C$

(4) $\int \frac{\ln(1-x^2)}{x^2\sqrt{1-x^2}}dx.$

8. (9 pts) Assume $f(x)$ is continuous on $[0,1]$ and differentiable on $(0,1)$. If $f(0)=f(1)=0$, $f(\frac{1}{2})=1$, prove that:

(1) there exists $c\in(\frac{1}{2},1)$, such that $f(c)=c$.

(2) For any real number k, there always exists $\xi\in(0,c)$, such that $f'(\xi)-k[f(\xi)-\xi]=1$.

11. 构造：$g(x)=f(x)-x$，则 $g(x)$ 在 $[0,1]$ 连续，$(0,1)$ 可导.

$g(\frac{1}{2})=f(\frac{1}{2})-\frac{1}{2}=\frac{1}{2}>0$　　$g(1)=f(1)-1=-1<0$

$g(\frac{1}{2})\cdot g(1)<0$　　故必存在一个 $c\in(\frac{1}{2},1)$，使得 $g(c)=0$ 即 $f(c)=c$，原命题得证.

14. 分析：关键是 构造函数 —— 解一个微分方程即可

草稿纸：$f'(x)-kf(x)+kx=1$

$f'(x)-kf(x)=1-kx$

$f(x)=\underbrace{\frac{e^{-\int -k\,dx}}{e^{kx}}}_{d}\left(\underbrace{\int(1-kx)e^{\int -k\,dx}}_{\downarrow}+C\right)$　　C 换成 F(x)，即要构造的函数

$\int(1-kx)e^{-kx}dx=\frac{1}{k}\int(kx-1)\,d(e^{-kx})=\frac{1}{k}(kx-1)e^{-kx}-\frac{1}{k}\int e^{-kx}d(kx)=$

$(x-\frac{1}{k})e^{-kx}+\frac{1}{k}e^{-kx}=xe^{-kx}$

故 $\frac{f(x)}{e^{kx}}=\frac{x}{e^{kx}}+F(x)$　得：$F(x)=\frac{f(x)-x}{e^{kx}}$

解：构造 $F(x)=\frac{f(x)-x}{e^{kx}}$，则 $F(x)$ 在 $[0,c]$ 连续，在 $(0,c)$ 可导.

$F(0)=\frac{f(0)-0}{e^0}=0=F(c)=\frac{f(c)-c}{e^{kc}}=0$

故根据罗尔定理，至少存在一个 $\xi\in(0,c)$ 使得 $F'(\xi)=0$

即 $\frac{[f'(\xi)-1]e^{k\xi}-ke^{k\xi}[f(\xi)-\xi]}{e^{2k\xi}}=0$

即 $f'(\xi)-k[f(\xi)-\xi]=1$，原命题得证.